Shortest Path with Negative Weights: Bellman Ford's Algorithm (ADA 2023, CSE 222)

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March 27, 2023

Dijkstra's Algorithm

Initialize $d[s] \leftarrow 0$;

For all $v \in V(G) \setminus \{s\}$, $d[v] \leftarrow \infty$;

$S \leftarrow \emptyset$ *and* $Q \leftarrow V(G)$;

while $Q \neq \emptyset$ **do**

  $u \leftarrow ExtractMin(Q)$;

  $S \leftarrow S \cup \{u\}$;

  for *each* $v \in N_G(u)$ **do**

    if $d[v] > d[u] + \text{weight}(u, v)$ **then**

      $d[v] = d[u] + \text{weight}(u, v)$;

      $\text{prev}(v) \leftarrow u$;

    end

  end

end

$Q = \{A, B, C\}$

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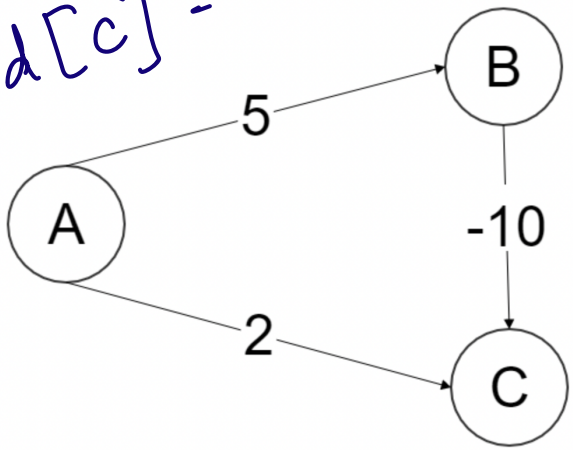
$Q = \{B\}$

$A = s$

$d[A] = 0$

$d[B] = 5$

$d[C] = 2$

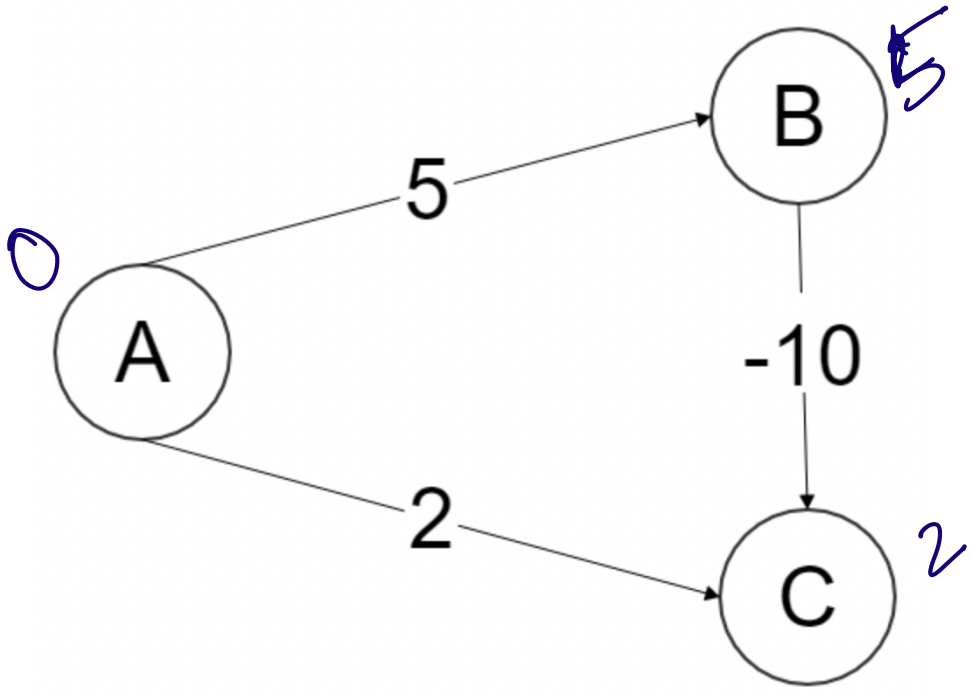


S

A B C

Presence of negative weighted edge

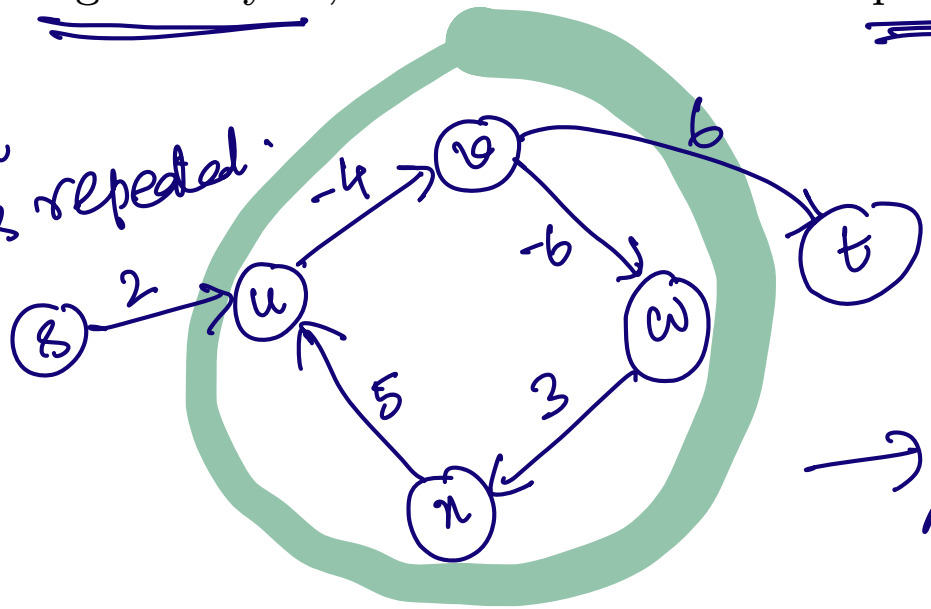
- **Input:** A directed graph $G = (V, E)$, every edge (u, v) has a weight$(u, v) \in \mathbb{R}$ and $s \in V(G)$.
- **Goal:** Paths from s to all the vertices of G with smallest total weight.

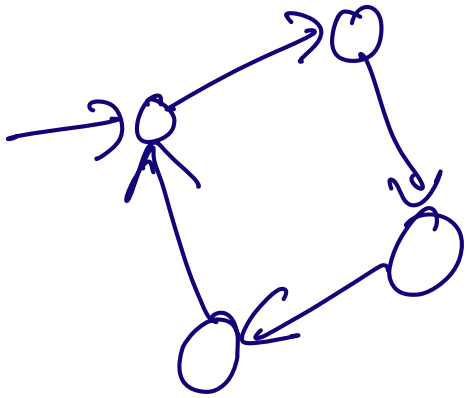


Outline

- **Assumption:** G has no negative cycle.
- If G has no negative cycle, then there is a shortest path from s to t that is simple.

Simple path
≡ no vertex is repeated.





Consider any directed cycle C.

$\sum_{(u,v) \in C}$ weight $(u,v) > 0$. Hence, using all the edges is of no use.

- If G has no negative cycle, then there is a shortest path from s to t that is simple.
- Let $OPT(i, v)$ is the minimum cost of a v-t path (a path from v to t) using at most i edges.
- Our original problem is to compute $OPT(n-1, s)$.
- **Lemma:** Let P be an optimal path from v to t. Then the following statements hold true.
  - If the path uses at most $i-1$ edges, then $OPT(i-1, v) = OPT(i, v)$.
  - If the path uses i edges and the first edge is (v, w), then $OPT(i, v) = OPT(i-1, w) + \text{weight}(v, w)$.

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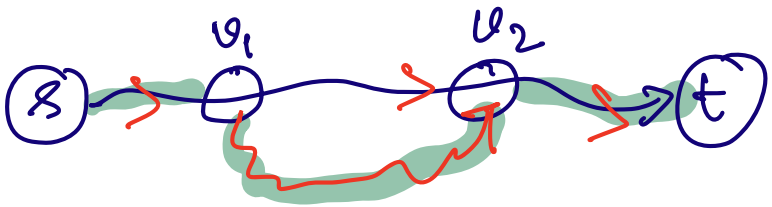
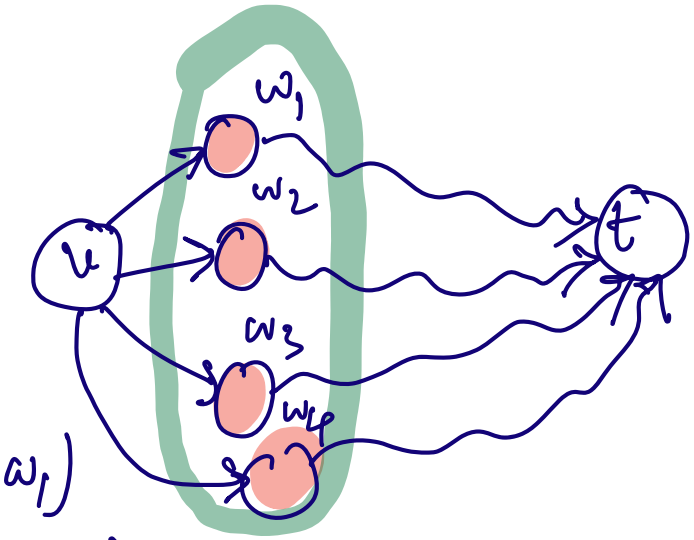
  - If the path uses at most $i-1$ edges, then $OPT(i-1,v) = OPT(i,v)$.
  - If the path uses i edges and the first edge is (v,w), then $OPT(i,v) = OPT(i-1,w) + \text{weight}(v,w)$.
- If $i > 0$, then

$$OPT(i,v) = \min\left\{OPT(i-1,v), \min_{w:\, w \in N(v)} \left(OPT(i-1,w) + \text{weight}(v,w)\right)\right\}$$

Shortest Path satisfies optimal substructure property.





$$\min \begin{cases} OPT(i-1, v) \\ OPT(i-1, w_1) + \text{weight}(v, w_1) \\ OPT(i-1, w_2) + \text{weight}(v, w_2) \\ OPT(i-1, w_3) + \text{weight}(v, w_3) \\ OPT(i-1, w_4) + \text{weight}(v, w_4). \end{cases}$$

For all $v \in V(G) \setminus \{t\}$, $M[0, v] \leftarrow \infty$;

$M[0, t] \leftarrow 0$;

for $i = 1, \ldots, n-1$ **do**

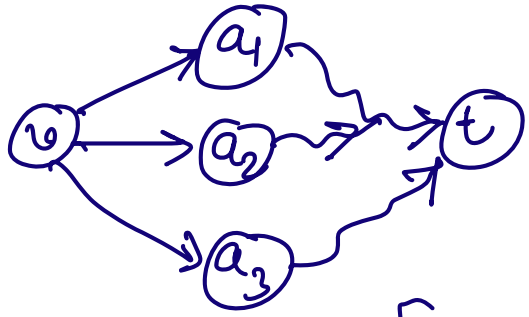
  for *each* $v \in V(G)$ *in any order* **do**

    Compute $M[i, v]$ *using the recurrence*;

  end

end

$M[n-1, s]$;



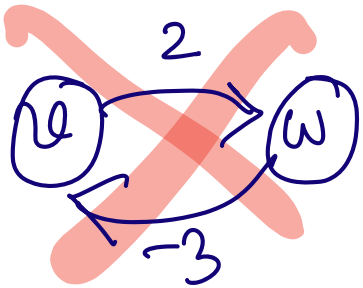
$$M[i,v] = \min \begin{cases} M[i-1, v] \\ M[i-1, a_1] + w(v, a_1) \\ M[i-1, a_2] + w(v, a_2) \\ M[i-1, a_3] + w(v, a_3) \end{cases}$$

Shortest path weight from s to t using at most $n-1$ edges.

For $i > 0$, $M[i, v] = \min \left\{ M[i-1, v], \min_{w : v \in N(w)} \left(M[i-1, w] + \text{weight}(v, w) \right) \right\}$

$w \in N(v)$

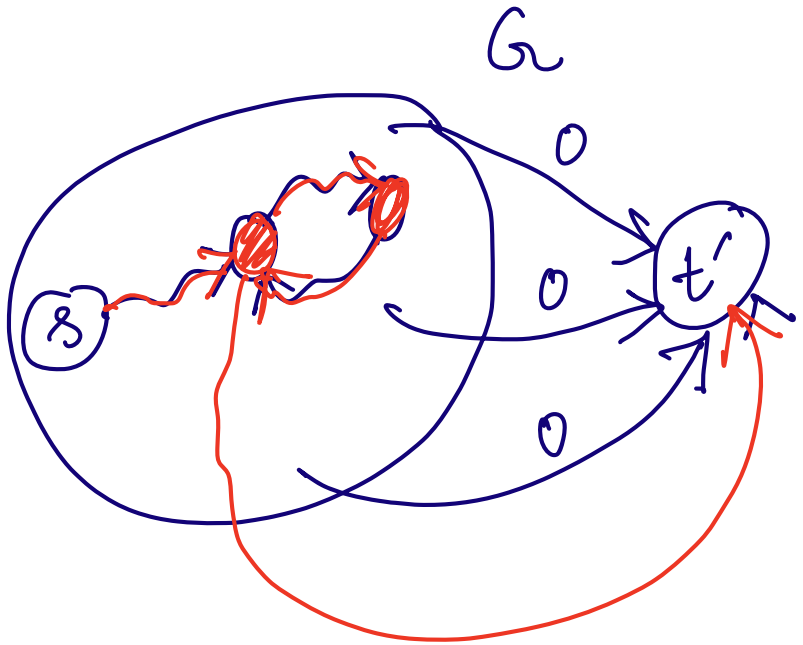
$M[1, v]$ then $M[2, v]$ ..... $M[n-1, v]$



$M[n, v]$

Detecting Negative Cycles

- **Augment the graph:** Add a new node t', and for all $v \in V(G)$, make (v, t) an edge. This graph is G'
- G' has a negative cycle C such that there is a path from C to t' if and only if G has a negative cycle.



- What will happen to $OPT(i, v)$ for all $i \geq n$ when there is no negative cycle in G?
- If there is a negative cycle in G, then how about what happens to the following quantity?

$$\lim_{i \to \infty} OPT(i, v)$$

- What will happen to $OPT(i, v)$ for all $i \geq n$ when there is no negative cycle in G? $OPT(n, v) = OPT(n-1, v) = OPT(n+2, v)$
- If there is a negative cycle in G, then how about what happens to the following quantity?

$$\lim_{i \to \infty} OPT(i, v)$$

$n = |V|$.

- If there is no negative cycle in G, then $OPT(i, v) = OPT(n-1, v)$ for all $i \geq n$.
- There is no negative cycle in G with a path to t if and only if for all vertices $v \in V(G)$, we have $OPT(n, v) = OPT(n-1, v)$.

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- If G has n nodes and $OPT(n, v) \neq OPT(n-1, v)$, then a path from v to t of total weight $OPT(n, v)$ contains a negative cycle.

Outline

Arbitrage

Suppose 1 USD has bought 0.82 EUR, 1 EUR bought 129.7 Japanese Yen, 1 Japanese Yen bought 12 Turkish Lira, and 1 Turkish Lira bought 0.0008 USD.

- **Input:** A collection of n currencies and their mutual exchange rates, e.g. $R[c_i, c_j]$ is the exchange rate from currency c_i to currency c_j.
- **Question:** Is there a sequence of currencies $c_i^1, \ldots, c_i^j$ such that the following holds true?

$$R[c_{i,1}, c_{i,2}] \cdot R[c_{i,2}, c_{i,3}] \cdots R[c_{i,j-1}, c_{i,j}] \cdot R[c_{i,j}, c_{i,1}] > 1$$

$c_{i,1}$ $c_{i,2}$ $c_{i,j}$

1 USD → 0.82 EUR → 0.82×129.7 Yen → $0.82 \times 129.7 \times 12$

1.03 USD ← $0.82 \times 129.7 \times 12 \times 0.0008$

$$\log\left(R[c_{i,1}\, c_{i,2}]\right) + \log\left(R[c_{i,2}\, c_{i,3}]\right) + \ldots + \log(c_{i,j}\, c_{i,1})$$

$$= \log\left(R[c_{i,1} c_{i,2}]\, R[c_{i,2} c_{i,3}] \ldots R[c_{i,j} c_{i,1}]\right)$$

$$> 0$$

$$\log \frac{1}{R} = -\log R$$

Take $\log\left(\frac{1}{R[c_i, c_j]}\right)$.